Experimentation Plan

1. Data Preparation & Preprocessing Experiments

Before training, you must establish the data pipelines for both streams.

- **RGB Stream (Stream A) Pipeline:** Process raw RGB video clips into sequences of 16 frames at a 224×224 resolution.

- **Skeleton Stream (Stream B) Pipeline:** Extract 17 body keypoints per frame using YOLO-Pose or MediaPipe. Convert these raw coordinate sequences into 3D Spatial-Temporal Heatmap Volumes to reduce sensitivity to sensor noise and poor lighting.

2. The Core Experiments & Baseline Comparisons

Your research design features a "Dual-Stream" experimental design that pairs generative Self-Supervised Learning (SSL) pipelines against established baselines.

Experiment A: RGB Stream vs. Traditional Supervised Baselines

- **Proposed Architecture:** Vision Transformer (ViT-Base and ViT-Large) acting as an asymmetric VideoMAE V2.

- **Baseline Architecture:** Traditional supervised 3D CNNs.

- **Pretraining Task:** Train the ViT on unlabeled Kinetics-400 or NTU RGB+D using Masked Video Reconstruction with a 90% Tube Masking ratio.

- **Comparison Goal:** Quantify the reduction in manually labeled training data required when utilizing generative SSL compared to the baseline traditional supervised methods that require massive, fully annotated datasets.

Experiment B: Skeleton Stream vs. Contrastive Learning Baselines

- **Proposed Architecture:** Spatial-Temporal Transformer (STH-MAE) utilizing dense 3D Spatial-Temporal Heatmap Volumes.

- **Baseline Architecture:** Graph Convolutional Networks (GCNs) operating on sparse, raw 3D joint/graph structures.

- **Pretraining Task:** Train the ST-Transformer on NTU RGB+D 120 (Skeleton data only) utilizing Masked Heatmap Reconstruction with random 3D masking.
- **Baseline Pretraining Task:** Contrastive Learning using InfoNCE optimization to maximize similarity between augmented views of the same action (positive pairs) and minimize similarity between different actions (negative pairs).
- **Comparison Goal:** Rigorously test if the generative heatmap representation offers the optimal "middle ground" of privacy and noise robustness compared to the contrastive baseline.

Experiment C: Cross-Modality Baseline Comparison

- **Goal:** Evaluate the performance trade-offs directly between Stream A (RGB video data) and Stream B (skeleton-based data).
- **Comparison Goal:** Assess the trade-offs in accuracy, computational latency, and privacy preservation between the pixel-level and geometric modalities.

Experiment D: Anomaly Detection Proxy Tasks & Domain Generalization

- **Proxy Tasks:** To bypass the need for exhaustive fall labels, integrate multi-task proxy learning during the fine-tuning stage. Specifically, train models to distinguish the "Arrow of Time" (forward vs. backward sequences) and "Motion Irregularity" (smooth vs. shuffled frames).
- **Domain Generalization:** Train the models on "staged" laboratory datasets (UP-Fall) and evaluate their ability to bridge the domain gap by testing them in "wild," real-world environments.
- **Testing Datasets:** Use GMDCSA-24 to test cross-domain generalization in varied home backgrounds, and OOPS-Fall as the "Gold Standard" to evaluate unpredictable real-world accidents.

Experiment E: Reference Stream Baseline

- **Proposed Architecture:** Reference stream utilizing MediaPipe or open-source object detection models for pose estimation, followed by classification via Graph Neural Networks (GNNs), LSTMs, and traditional ML classifiers.
- **Comparison Goal:** Establish a baseline for pose-based detection and compute

downward/fall velocity as a primary point of comparison against more complex generative models.

3. Evaluation Metrics & Efficiency Tracking

Standard accuracy metrics will be discarded due to extreme class imbalance (99% non-fall standard activities). To evaluate the technical feasibility and address alert fatigue, compute the following:

- **Sensitivity & Specificity:** Crucial for ensuring true falls are caught while avoiding missed alarms.
- **F1-Score:** The harmonic mean to balance precision and recall on imbalanced data.
- **False Alarm Rate (FAR):** Critical clinical metric to reduce caregiver "alert fatigue".
- **Computational Latency (FPS):** Critical to ensure the system architecture is feasible for real-time edge devices.

Phase / Component	**Stream A (RGB / Pixel-Level)**	**Stream B (Skeleton / Geometric)**	**Supervised RGB Baselines**	**Contrastive Skeleton Baselines**
Data Modality	Raw RGB Video Clips (224×224, 16 frames)	3D Heatmap Volumes (17-Keypoint via YOLO/MediaPipe)	Raw RGB Video Clips	3D Joint Coordinates/Graphs
Core Architecture	Vision Transformer (ViT-Base & ViT-Large) / VideoMAE V2	Spatial-Temporal Transformer / STH-MAE	Traditional 3D CNNs	Graph Convolutional Networks (GCNs)

Pretraining Dataset	Kinetics-400 or NTU RGB+D (Label-free)	NTU RGB+D 120 (Skeleton only)	N/A (Supervised reliance)	NTU RGB+D 120 (Skeleton only)
Pretraining Objective	Masked Video Reconstruction (90% Tube Masking)	Masked Heatmap Reconstruction (Random 3D Masking)	Supervised Action Recognition	Contrastive Learning (InfoNCE Optimization)
Fine-Tuning Dataset	UP-Fall (Staged Multi-Camera Falls)	UP-Fall (Staged Multi-Camera Falls)	UP-Fall	UP-Fall
Proxy Tasks (Fine-Tune)	Arrow of Time, Motion Irregularity	Arrow of Time, Motion Irregularity	None	None
Generalization Testing	GMDCSA-24 & OOPS-Fall ("Wild" sets)	GMDCSA-24 & OOPS-Fall ("Wild" sets)	GMDCSA-24 & OOPS-Fall	GMDCSA-24 & OOPS-Fall
Primary Evaluation Focus	Latency vs. Accuracy	Noise-robustness & Privacy	Data Labeling Bottleneck	Viewpoint-Invariance